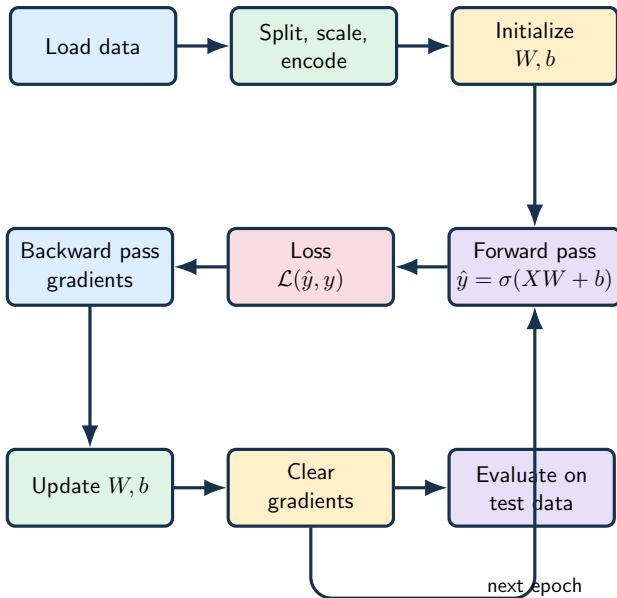
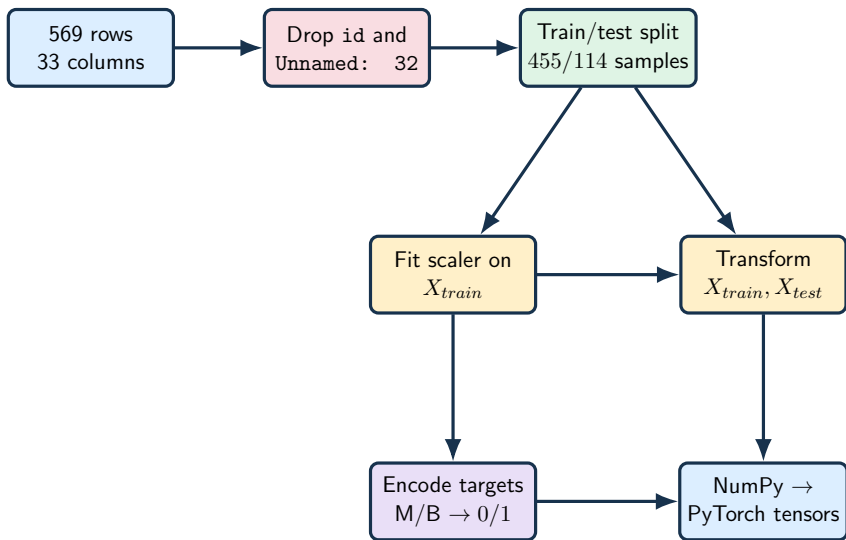


End-to-end training pipeline

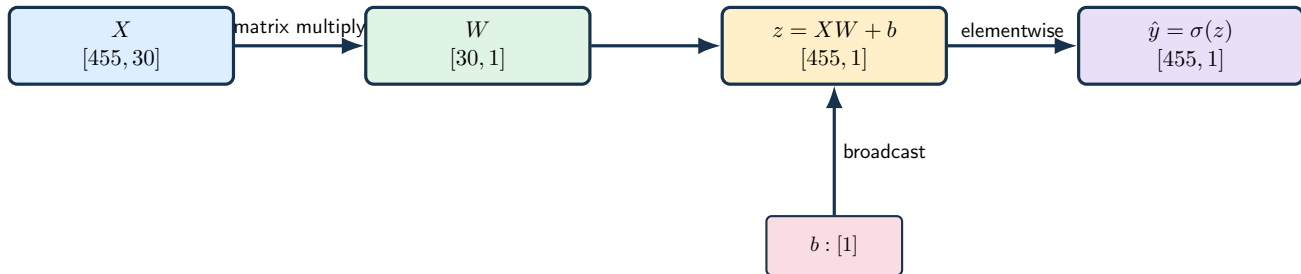


Data preparation and leakage-safe scaling



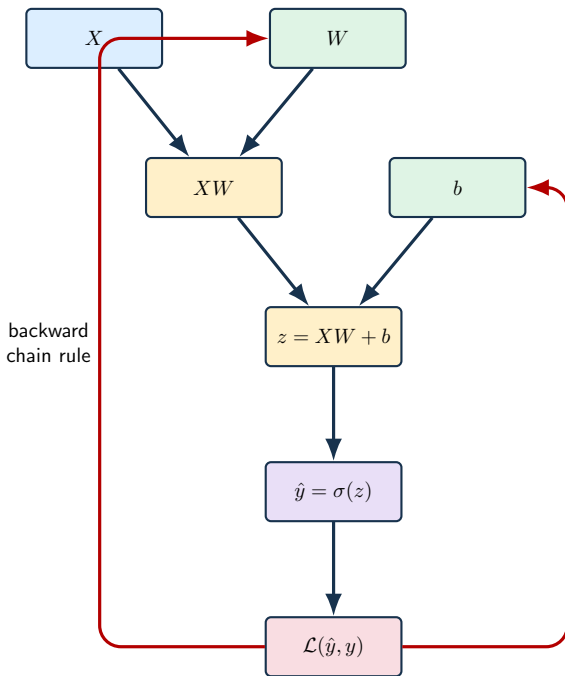
Only training statistics should determine the scaler. The target must be numeric, and feature/parameter dtypes must match.

Single-neuron tensor-shape flow



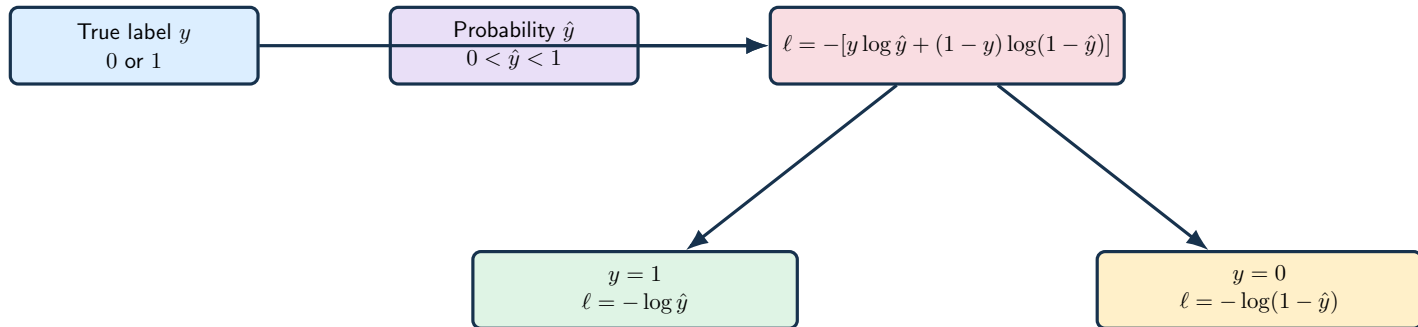
Thirty input features require thirty weights. One output neuron contributes one bias and one probability per sample.

Computational graph and gradient paths



`loss.backward()` follows the graph and stores $\partial\mathcal{L}/\partial W$ and $\partial\mathcal{L}/\partial b$ in `.grad`.

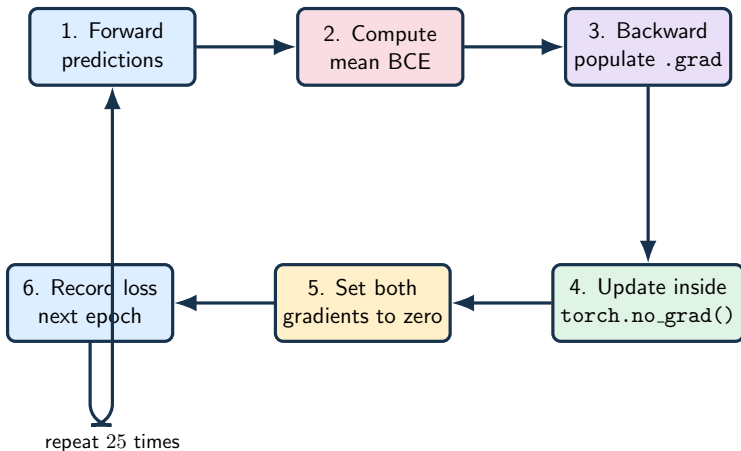
Binary cross-entropy for one sample



A confident wrong prediction receives a large penalty.

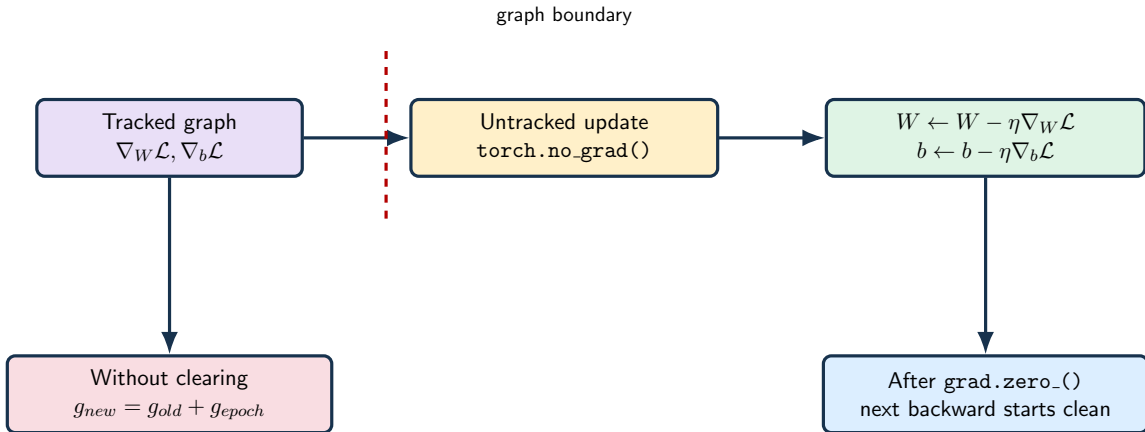
Clamping avoids evaluating $\log 0$ numerically.

One epoch: forward, backward, update, clear

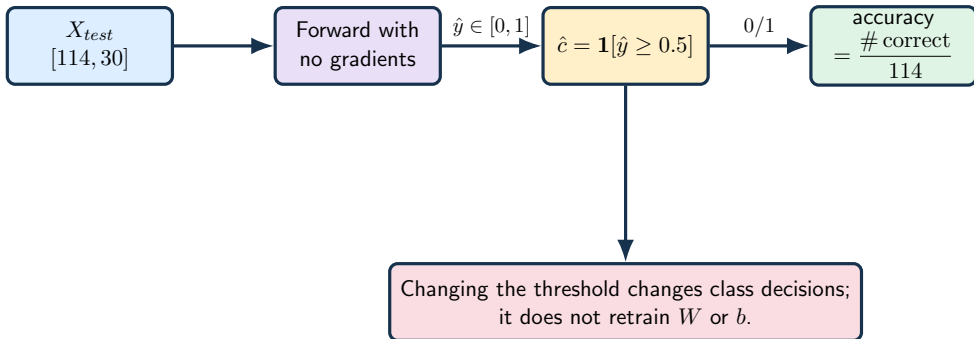


PyTorch accumulates gradients by default. Clearing them is required before the next backward pass.

Parameter update and graph boundary



Evaluation and thresholding



Accuracy is only one metric. Precision, recall, and F1 are often important for medical classification.